

FIKIRU DEMLEW

Full-Stack Developer · Frontend Specialist · ML Engineer

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PROFILE

Graduated Mechatronics Engineer and self-driven full-stack developer with hands-on experience across the entire product lifecycle — from pixel-perfect React and Vue frontends to Go and Node.js backends, GraphQL via Hasura, and containerised deployments. Has delivered real-world projects for private remote clients and led frontend teams under GDSC. Complements engineering skills with applied machine learning: trained CNN image classifiers with MobileNetV2 and built advanced segmentation pipelines using OneFormer and SAM. Eager to join a team where breadth of stack and engineering rigour both matter.

TECHNICAL SKILLS

Languages	JavaScript (ES6+), TypeScript, Python, Go, C++
Frontend	React, Next.js, Vue 3, HTML5, CSS3, Tailwind CSS
Backend / API	Node.js, Go (Gin), Hasura (GraphQL), REST, Swagger / OpenAPI
Databases	PostgreSQL, Firebase / Firestore
DevOps & Tools	Docker, Postman, Git, GitHub, Netlify, Vercel
Machine Learning	TensorFlow, Keras, CNN, MobileNetV2, OneFormer, SAM-3, Transfer Learning
Payments	Payment gateway integration (e-commerce)
Soft Skills	Team leadership, Remote collaboration, Communication, Adaptability

WORK EXPERIENCE

Freelance Full-Stack Developer · [Private Clients \(Remote\)](#) 2025 – Present

- Delivered full-stack web applications for private clients on a remote, contract basis.
- Handled end-to-end scope: requirement gathering, UI/UX design, development, and deployment.
- Worked independently across multiple stacks depending on client needs.

PROJECTS

YouTube Clone · [Personal Project](#) 2024

- Built a fully functional YouTube clone with video browsing, search, and category filtering powered by the YouTube Data API v3.
- Implemented responsive layout closely mirroring YouTube's UI, with dynamic data fetching and React state management.
- [Live at youtubeclone1994.netlify.app](https://youtubeclone1994.netlify.app)

Stack: React · Tailwind CSS · YouTube Data API v3 · Netlify

Full-Stack Food Recipe Sharing Platform · [Personal Project](#) 2024–2025

- Designed and built a full-stack recipe sharing site where users can create, browse, and share recipes with rich media.

- Architected a Go backend with Hasura (GraphQL) for flexible, real-time data querying and mutations over PostgreSQL.
- Containerised the entire stack with Docker for consistent local development and deployment; documented APIs with Swagger and tested endpoints with Postman.
- Delivered a responsive, component-driven frontend in Vue 3 styled with Tailwind CSS.

Stack: *Vue 3 · Tailwind CSS · Go · Hasura (GraphQL) · PostgreSQL · Docker · Swagger · Postman*

E-Commerce Application with Payment Integration · [Personal Project](#)

2025

- Developed a full-featured e-commerce web app with product listings, cart management, order flow, and secure checkout.
- Integrated a third-party payment gateway, handling transaction lifecycle, success/failure callbacks, and order confirmation.
- Built a PostgreSQL-backed REST API to manage products, users, and orders; implemented auth with JWT.

Stack: *React · PostgreSQL · Node.js · Payment Gateway API · REST API · JWT*

AI Tile Visualizer — Semantic Segmentation · [ML Project](#)

2025

- Built an AI pipeline that segments room surfaces (walls, floors, ceilings) in photos and realistically composites new tile textures onto them.
- Used OneFormer for panoptic/semantic segmentation to identify surface regions with high accuracy across diverse room layouts.
- Integrated SAM-3 (Segment Anything Model) for precise, user-guided mask refinement on complex surface boundaries.
- Delivered a Python pipeline taking a room image and a tile texture as input and producing a visualised output image.

Stack: *Python · OneFormer · SAM-3 · TensorFlow / Keras · OpenCV · Pillow*

Tomato Sorting AI — CNN + Transfer Learning · [ML Project](#)

2024

- Trained a CNN image classifier using MobileNetV2 transfer learning to automatically grade and sort tomatoes by quality.
- Applied fine-tuning and data augmentation to adapt the pre-trained backbone to a custom tomato dataset, achieving high classification accuracy.
- Gained end-to-end ML pipeline experience: data preparation, augmentation, training, evaluation, and export.

Stack: *Python · TensorFlow · Keras · MobileNetV2 · CNN · Transfer Learning*

Delivery App — Frontend · [Google Developer Student Clubs \(GDSC\)](#)

2023

- Collaborated in a 9-member cross-functional team to build a production-ready, mobile-optimised delivery app interface.
- Developed key UI components ensuring smooth integration with a Go backend via REST APIs.

Stack: *React · Tailwind CSS · JavaScript · PostgreSQL · Go*

Movie Watchlist Platform · [Google Developer Student Clubs \(GDSC\)](#) — Team Leader (5 members)

2022–2023

- Led a 5-member team to build a web platform aggregating trending movies globally, managing full project workflow from idea to deployment.
- Implemented real-time search and a personalised watchlist backed by Firebase authentication and Firestore.

Stack: *HTML · CSS · JavaScript · React · Firebase*

EDUCATION

B.Sc. Mechatronics Engineering · [Addis Ababa Science and Technology University \(AASTU\)](#)

2021 – 2026

- Multidisciplinary programme covering control systems, embedded systems, robotics, and software engineering.
- Active GDSC member — Frontend Developer & Team Leader.

Programming Fundamentals Nanodegree — Udacity

Verified Certificate · Python, problem-solving, and algorithmic thinking.

Machine Learning Course

Covered supervised learning, CNNs, and transfer learning; applied knowledge in tomato-sorting and tile-visualizer projects.

Natural Science Stream — Debreselam St. Mary School, Gondar